

SHIVAM KOTAK

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EDUCATION

University of Southern California, Los Angeles, CA

January 2022-December 2023

Master of Science in Computer Science

CGPA:3.71

Relevant Coursework: Web Technologies, Database Systems, Information Retrieval (in-progress), Foundations of AI, Applied NLP, Machine Learning, Analysis of Algorithms

University of Mumbai, Mumbai, India

July 2021

Bachelor of Engineering in Computer Engineering

CGPA:9.08/10

EXPERIENCE

CCC Intelligent Solutions, Chicago, United States

August 2023-Present

Data Science Intern

- Leveraged **Vision Language Models** and **Visual Question answering** models to streamline the process of subrogation in auto insurance claims. Used AWS (Sagemaker, S3, Lambdas, EC2) to test and deploy models.

Cloud Counselage Pvt Ltd, Mumbai, India

March 2020-July 2020

Machine Learning Intern

- Developed an Event Recommender System by extracting domain and type of an event such as seminars, hackathons, expos, etc., from the event headline and selects employees to whom an event can be recommended to.
- Incorporated concepts of **TF-IDF**, **lemmatization** and **linear support vector classifier** to build the model

TECHNICAL SKILLS

- Programming Languages:** Python, C, C++, Java | **Web Development Stack:** HTML, CSS, Vanilla JS, React JS, Angular, Typescript, PHP, Flask, AWS, GCP, Node.js, Android, Express.js, Bootstrap, Material UI | **Database:** SQL, Oracle SQL, PostgreSQL, MongoDB
- Machine Learning/Data Science:** PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, SpaCy, NLTK, OpenCV, AWS (Sagemaker, S3, EC2, Lambda), MLFlow, Scipy, Numpy, Pandas, Transformers, Gen AI, Matplotlib, Seaborn, Optuna
- Frameworks and Tools:** JavaFX, PowerBi, Git, Docker, Hadoop, Bash, Linux

ACADEMIC PROJECTS

Stock Price Prediction Dashboard – [Link to Repo](#) – Python, TensorFlow, Flask, React.js

- Predicted end-of-day share price for a company on the National Stock Exchange. Leveraged the **Keras** library in Python by employing a **CNN-LSTM** neural network. Employed **React.js**, **Material UI** and **chart.js** for the front end.
- Collated historical data using a python library and evaluated data based on technical indicators such as **RSI**, **MACD** etc
- Trained a prediction model based on past 4-5 years data. Predicted values within a variation of approximately **10-15%** of the actual price.

Manga Downloader – [Link to Repo](#) – Java, JavaFX, Web Scraping, HTML Scraping, Selenium, BeautifulSoup4

- Developed an application that takes input as a name of a Japanese comic and then lists and downloads chapters of a comic
- Coded in **Java**, leveraging **JavaFX** for the UI. It works by parsing the HTML of a source website (mangadex.org) for identifying, listing, and downloading chapters. The pages were scraped by creating a selenium session to read the HTML of dynamically loaded web pages.

Football Commentary Guided Player Rating Prediction – [Link to Repo](#) – Python, PyTorch

- Explored the use of **BERT** and **XLNet** to encode commentaries to test on different models (Both Classical and Deep Models) with Negative Log Likelihood loss and MSE loss. The best result loss of **0.35** was achieved with **XLNet** and an **LSTM** model.
- Gathered data by collating from multiple datasets and API. Final processed dataset can be found on Kaggle at: [Football Player Rating, Stats and Event Commentary](#)

Event Search Website – [Link to Webpage](#) – Angular, Typescript, HTML, CSS, Bootstrap, JavaScript, NodeJS

- Created a **responsive** single-page web application to search for events using the TicketMaster API. The webpage allows the user to search for events through keywords, distance, manual location as well as location auto-detection.
- The webpage gives detailed event and venue information including seat map and ticket-status, artist information, location, genre, time and date, and relevant links and also allows users to favorite events and store those events in **LocalStorage**.

CERTIFICATIONS

- Generative Adversarial Networks (GANs) Specialization by DeepLearning.AI via Coursera
- Applied Plotting, Charting & Data Representation in Python by University of Michigan via Coursera

LEADERSHIP AND INVOLVEMENT

- Robotics Instructor and Program Coordinator for the USC Information Sciences Institute x Amazon Stimulating Stem Summer Program for inner city high school students in Los Angeles
- Secured 1st Place at the college round of the National Smart India Hackathon, 2020. Worked with the team to build a prototype of an autonomous drone delivery system using various technologies such as python, android etc.